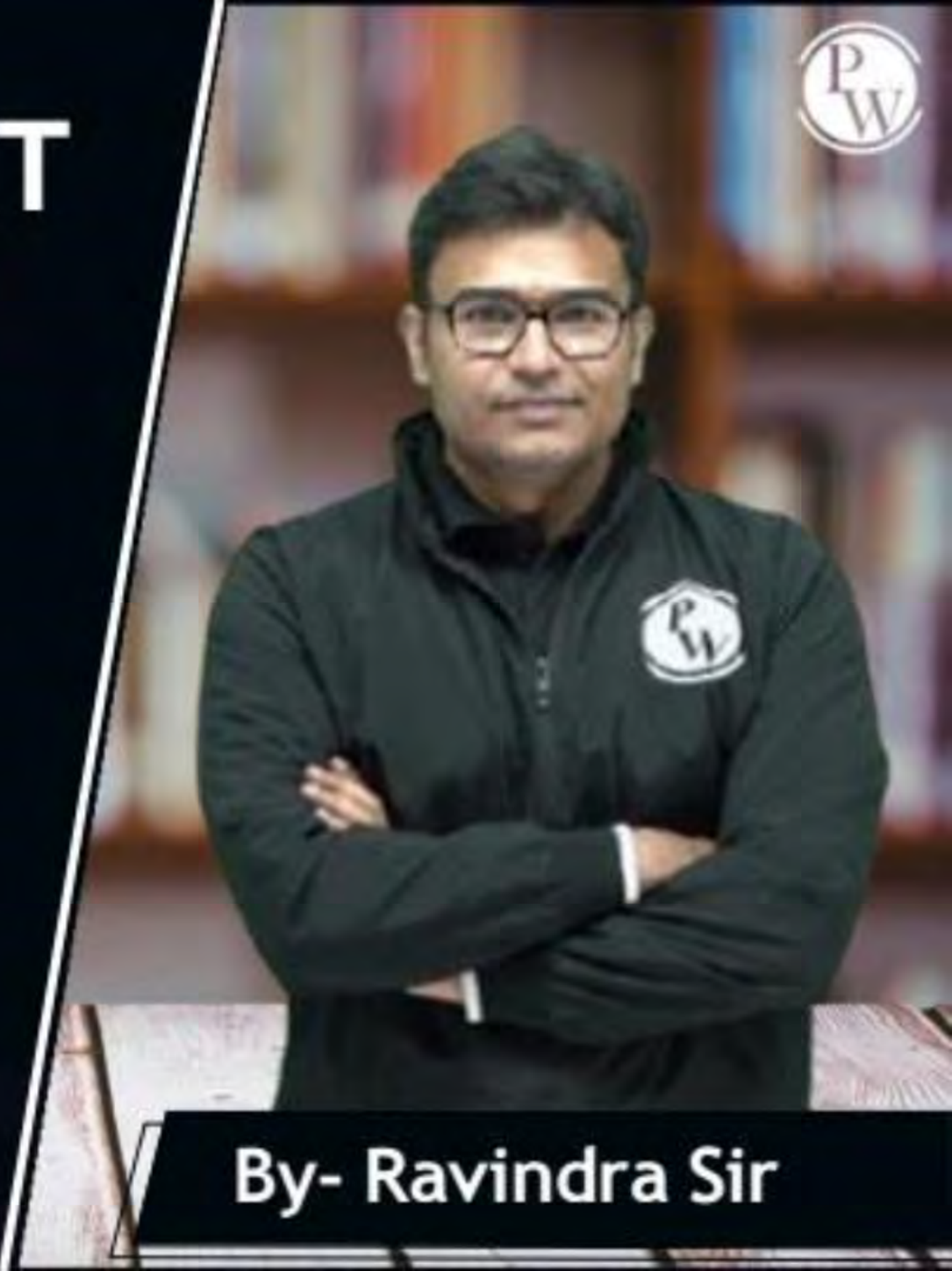
Computer Science & IT

COMPUTER NETWORK
(CN)

Routing

Lecture No. 02



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

DVR

Topics to be Covered



- Topic
- Topic
- Topic

LSR

Inspiring Stories : Bahia Bakari



Background: French 12-year-old, flew from Paris.

Struggle: Plane crashed into the ocean, no life jacket.

Achievements: Clung to debris, floated alone for 9 hours until rescue.

Impact: Called “miracle girl” for her survival.

Inspiring Stories : Poon Lim



Background: Chinese sailor, WWII era.

Education: Ship sank in South Atlantic, left alone on a raft.

Achievements: Survived 133 days at sea using makeshift tools, fishing, and rainwater.

Impact: Holds record for longest survival on a raft.

Inspiring Stories : Aron Ralston



Background: American climber.

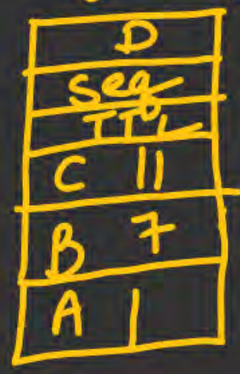
Education: Arm pinned by boulder in remote canyon, alone.

Achievements: After 5 days, amputated his own arm, then hiked out.

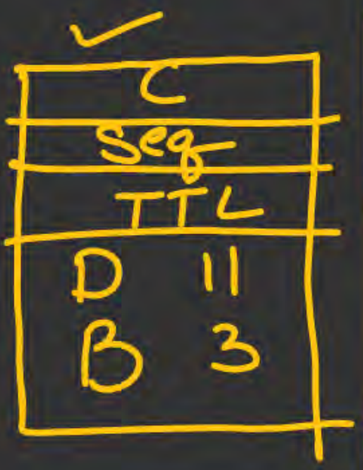
Impact: Escaped death through sheer will.

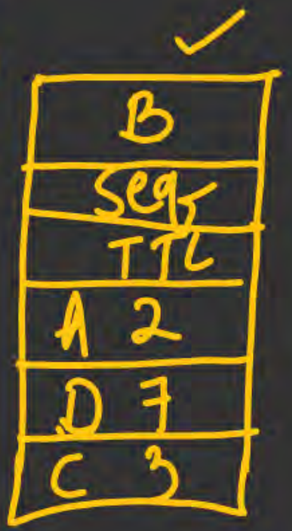
Link State Routing :-

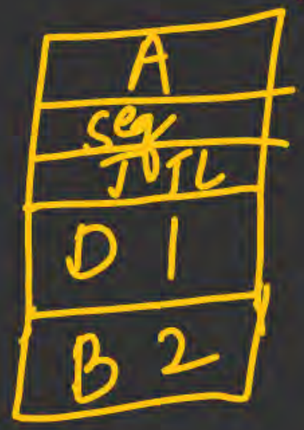
Step 1 :- Every Router prepares Link state packets





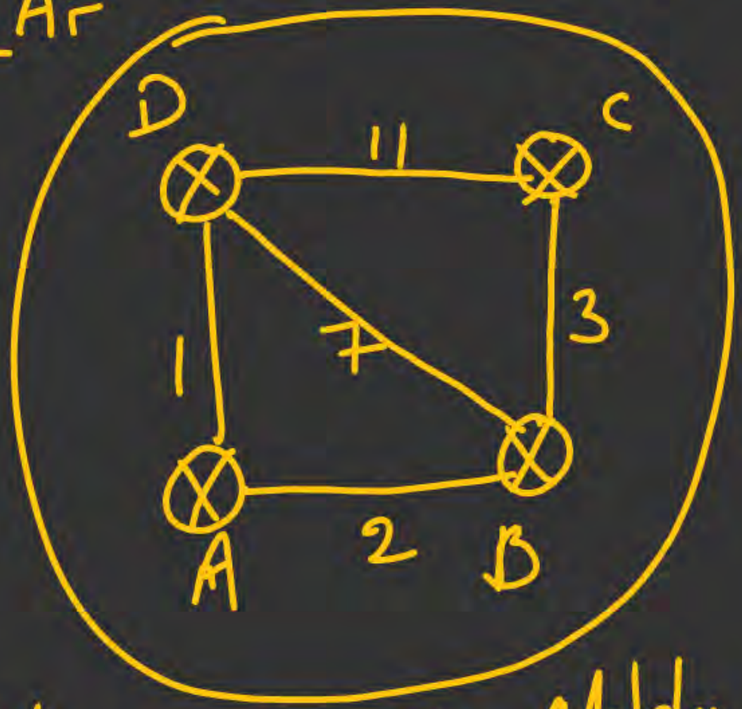






Step 2 :- Every Router will flood the LSP to every other Router

At A :-



Step 3: A will apply Dijkstra algo.

DVR is based on local knowledge

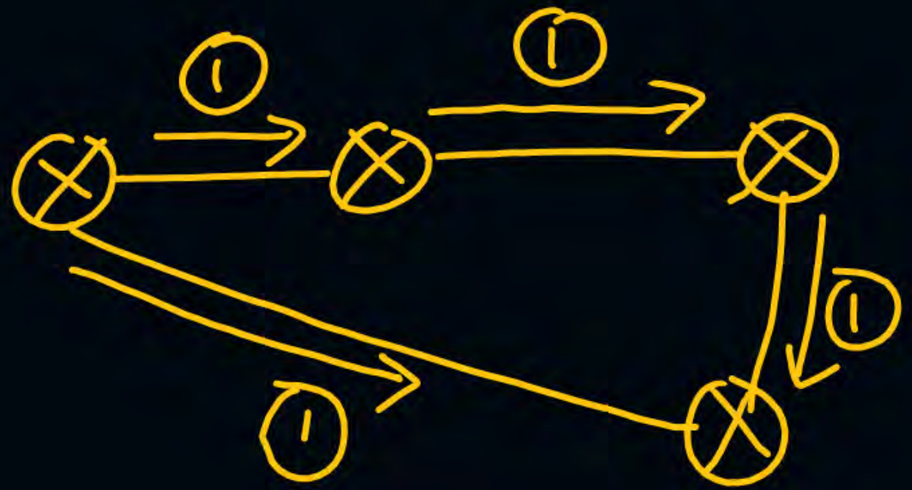
LSR is based on global knowledge

LSR converges fast

Disadv:- Flooding is used in LSR.

Heavy traffic

Duplicate packets


1
1
1
1

Seq numbers are used to find out which packet is latest.

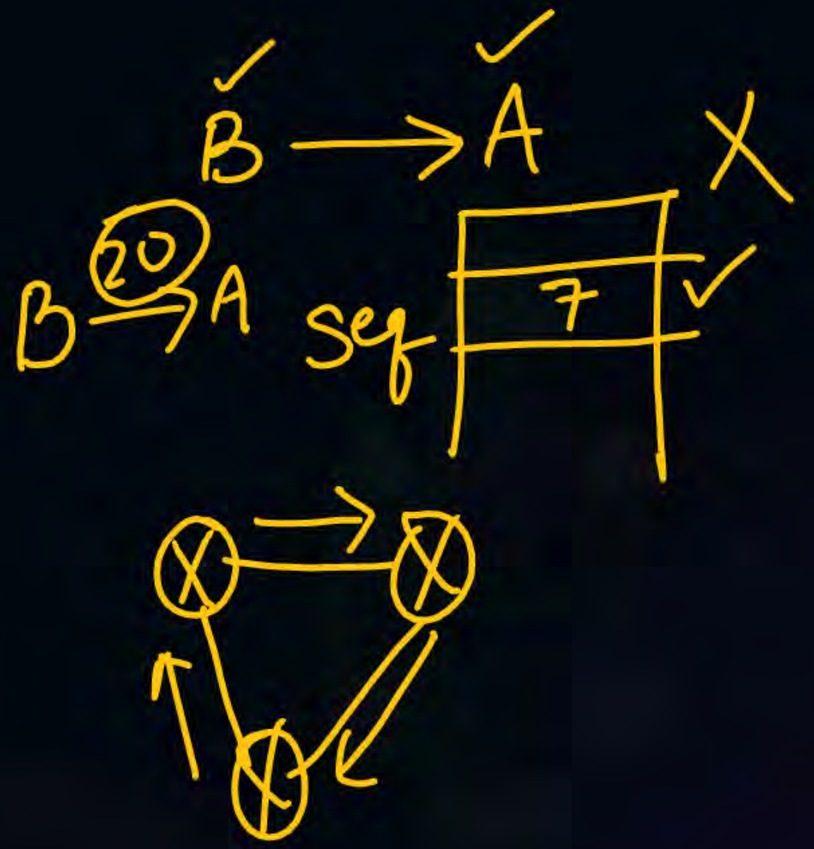
Every Router will maintain a DB

At A:

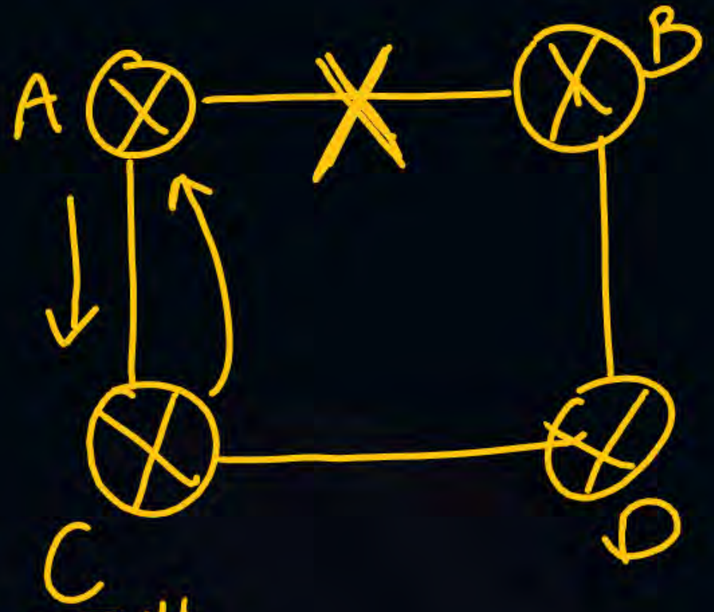
Router	Seq	Life time
B	~~20000~~	~~5 min~~
B	20	5 min
C	30	5 min
D		

B → A (11) 1000

11 X
12 X
13 X
14 X
⋮
1000 X



TTL: To avoid infinite loop





A → B

Transient problem

→ Temporary problem

loops — Trans

DVR :- loops → persistent problem

↓

Permanent

RIP ← DVR

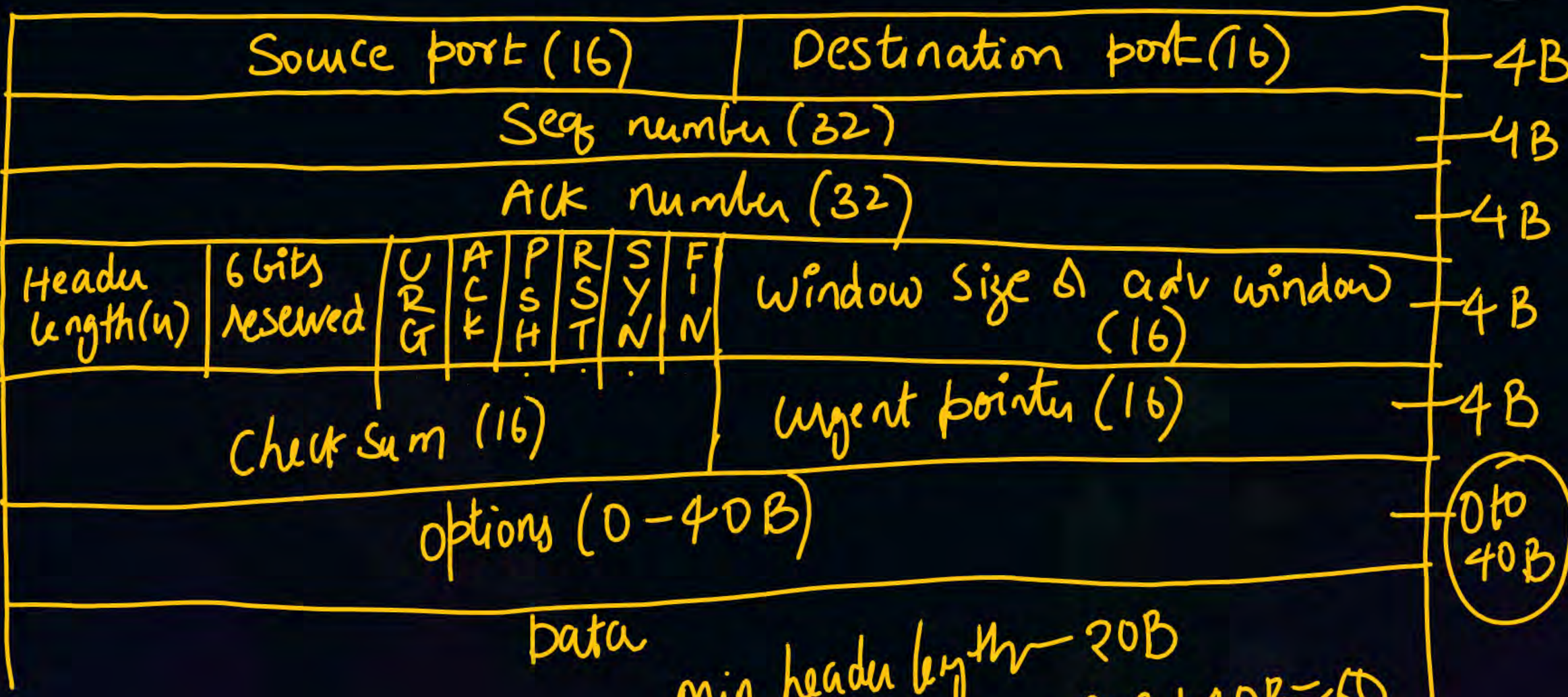
X

1) 1980's
2) BW required is less
3) Based on local knowledge
4) Bellman ford
5) Traffic is less
6) Periodic updates
7) Converges slow
8) Count to ∞
9) Persistent loops

LSR → OSPF X

1) 1990
2) BW required is high
3) Based on global knowledge
4) Dijsktra
5) Traffic is high
6) Periodic
7) converges fast
8) NO Count to ∞
9) Transient loop

TCP header



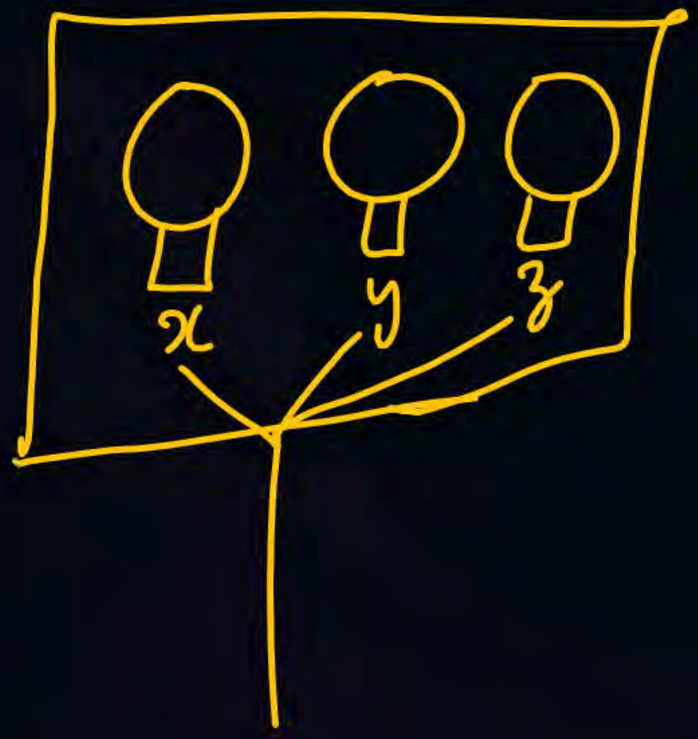
min header length - 20B

max header length - 20B + 40B = 60

SP and DP :- (16 bits) (0 - 65,535)

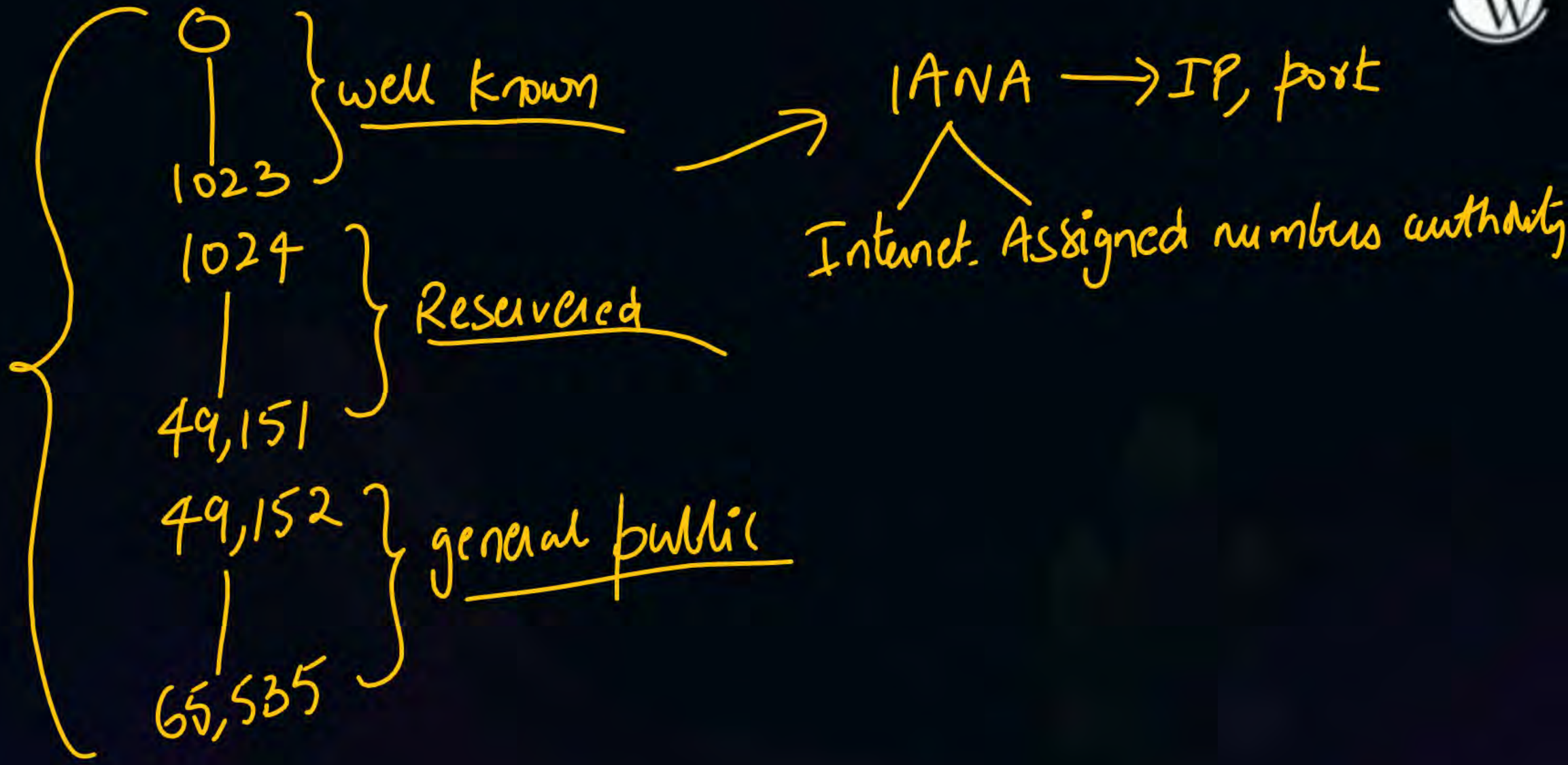
TCP is a end to end protocol

Port numbers are used for multiplexing and Demul



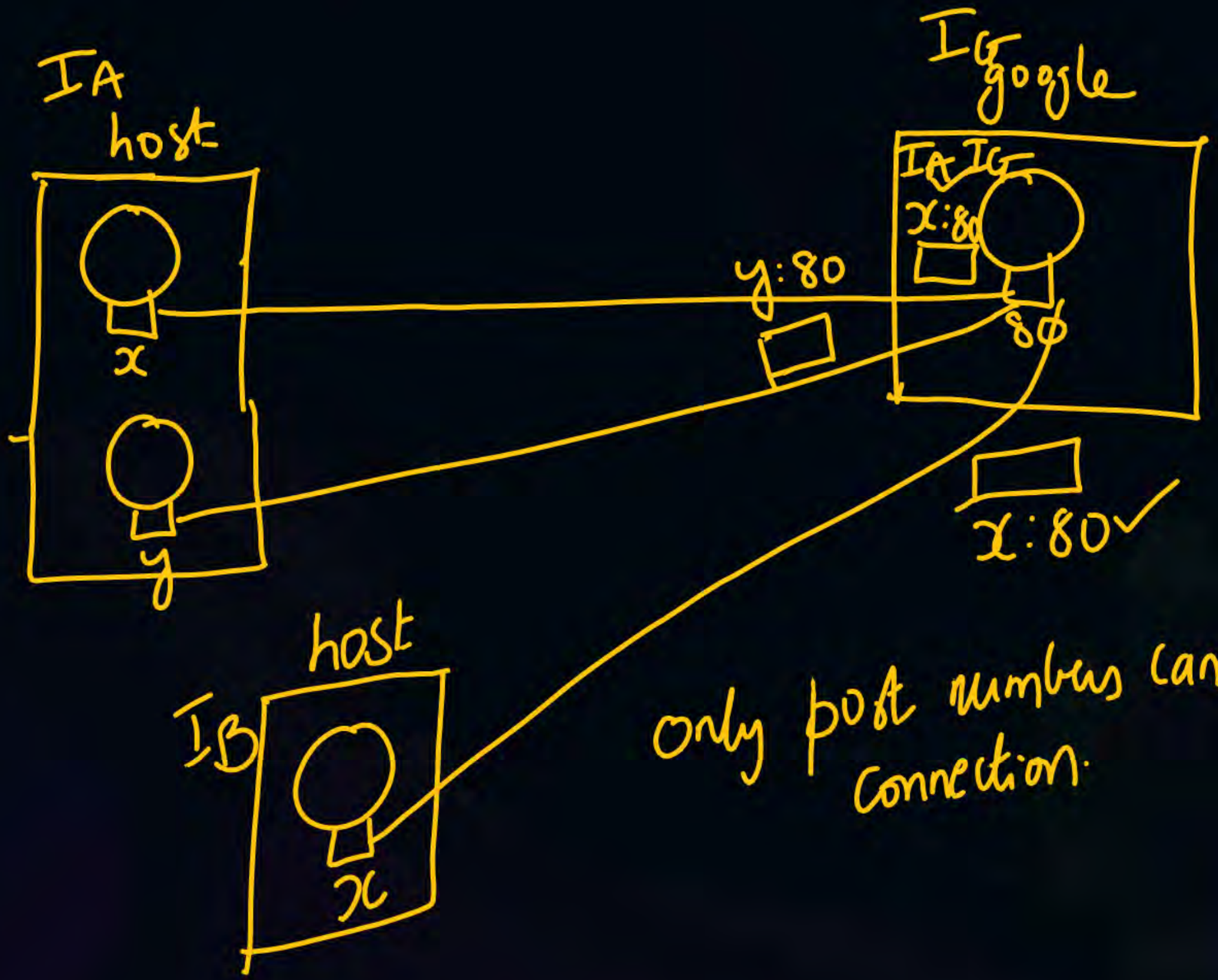
All well known services have fixed port numbers

HTTP: 80
FTP-21
Telnet -23
SMTP-25


0
1023
well known
IANA → IP, port
Internet Assigned numbers authority
1024
49,151
Reserved
49,152
65,535
general public

TCP is a connection oriented protocol.

Before sending the data, resources (BW, CPU, Buffer) are reserved



only port numbers cannot identify a connection.

IP + port → Can uniquely identify a connection
32 16

↓

Socket

(32 + 16) = 48 bits

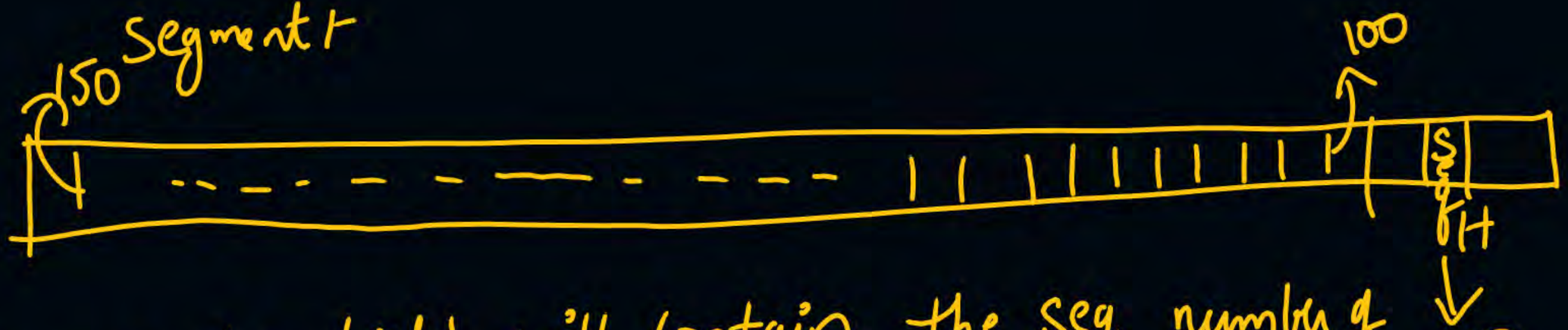
Sequence number :-

TCP is byte stream protocol → every byte is numbered

IP is a packet stream protocol → every packet is number (Identification no)

DLL – HDLC → bit stream

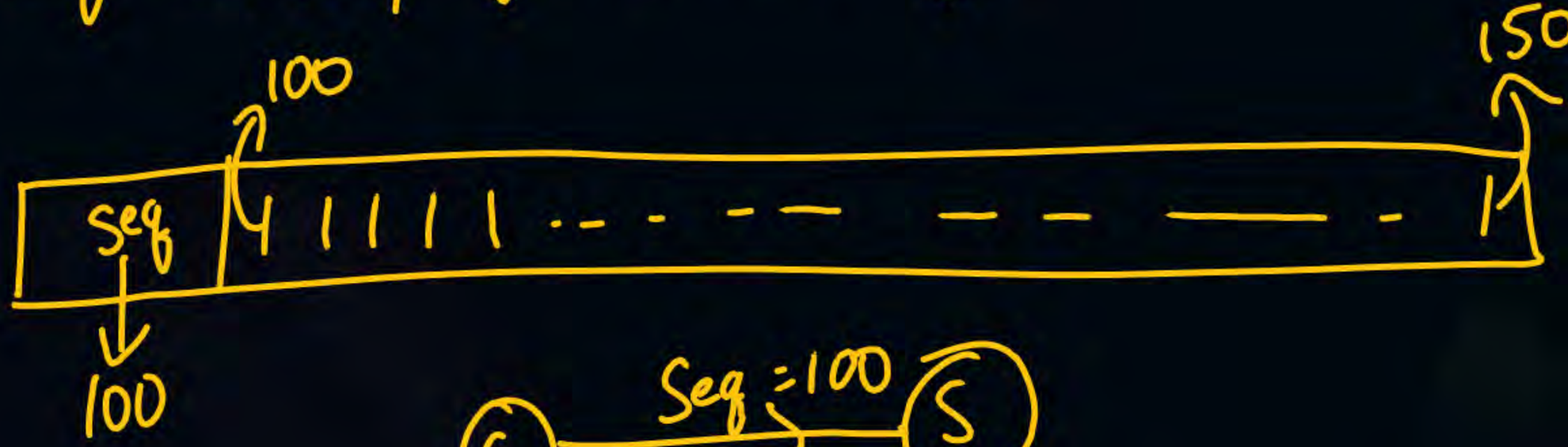
X



Seq number field will contain the seq number of 100 1 byte of the data

Acknowledgment number :-

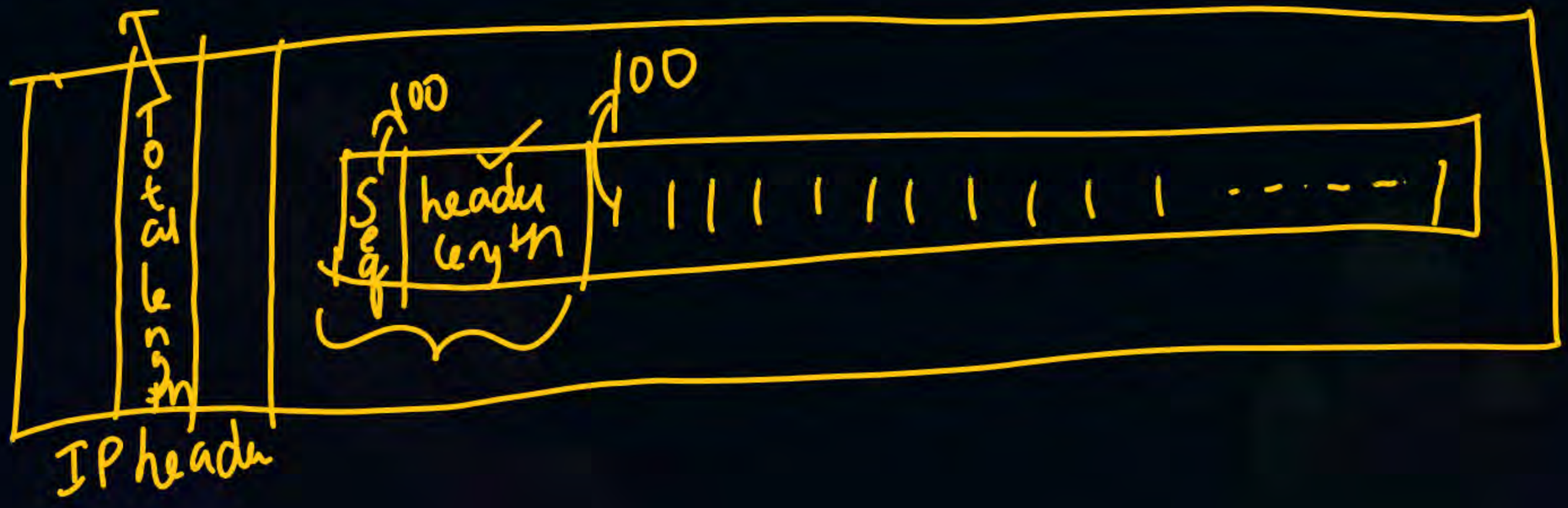
The Seq number of the the byte that is expected next



ACK number = last byte Seq number + 1.

How will a process know the last byte seq number?

TCP header → Total length X

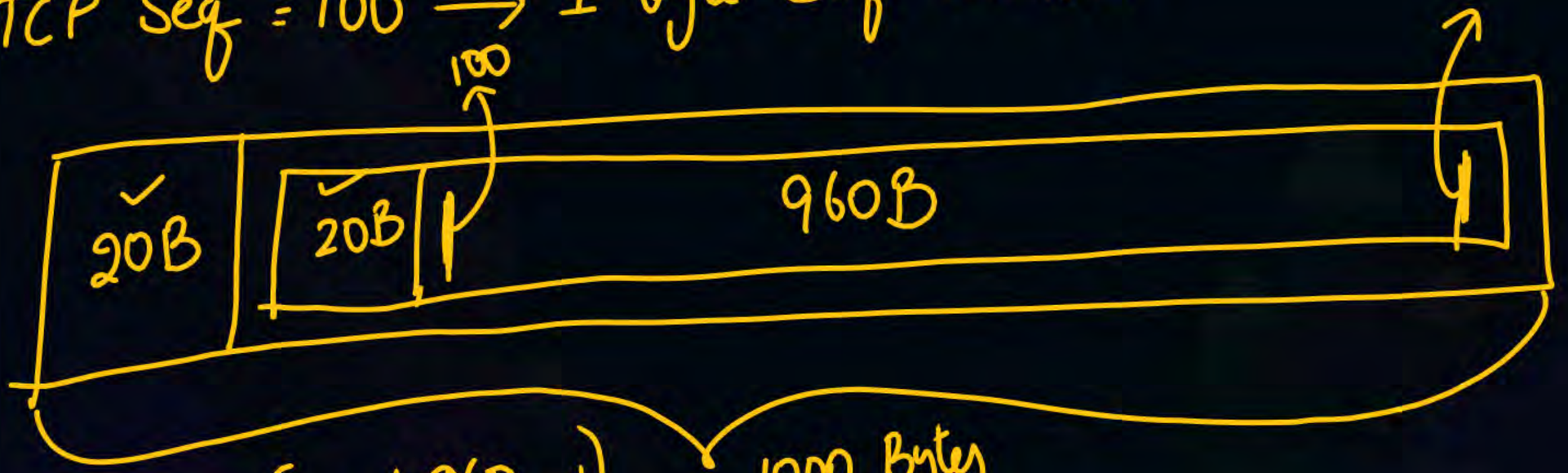


IP datagram total length = 1000B

IP header length = 5 × 4 = 20

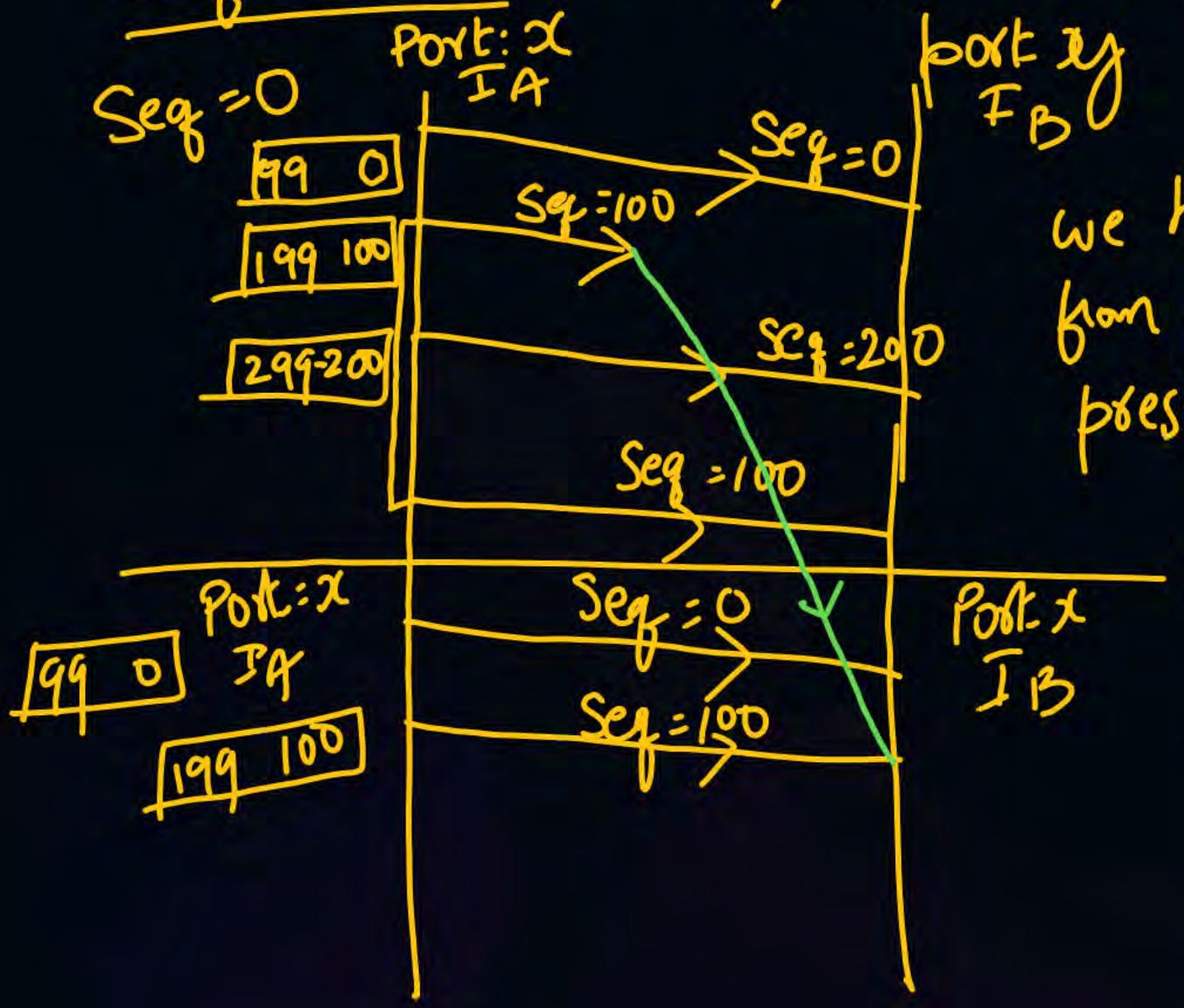
TCP header length = 5 × 4 = 20

TCP Seq = 100 → 1 byte seq number



Last byte :

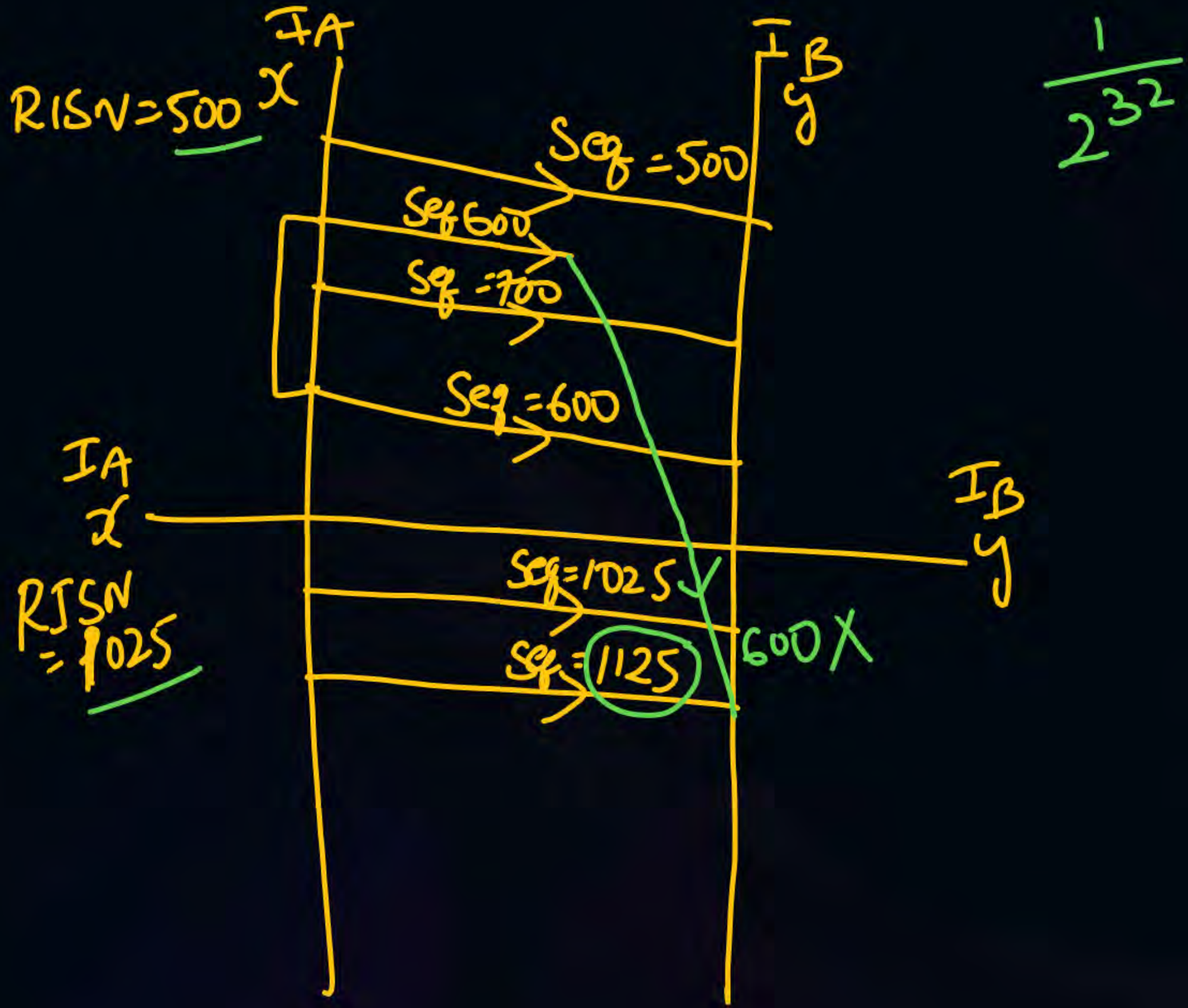
Seq number (32 bits) $(0 - 2^{32} - 1)$



we have to avoid accepting packet from previously closed connection in the present connection. So we use RISN.

Connection close

LT of a packet in internet is 3 min.


IA
X
IB
y
RISN=500
Seq=500
Seq=600
Seq=700
Seq=600
IA
X
IB
y
RISN
=1025
Seq=1025
Seq=1125
600 X

$\frac{1}{2^{32}}$



Wrap around :- using up all seq numbers and repeating the sequence number is called wrap around.

Wrap around time :- Time taken to repeat the same seq number.

WAT depends on the BW

2^{32} → Seq num

BW - 1MBps

10^6 B — 1 sec

10^6 Seq num — 1 sec

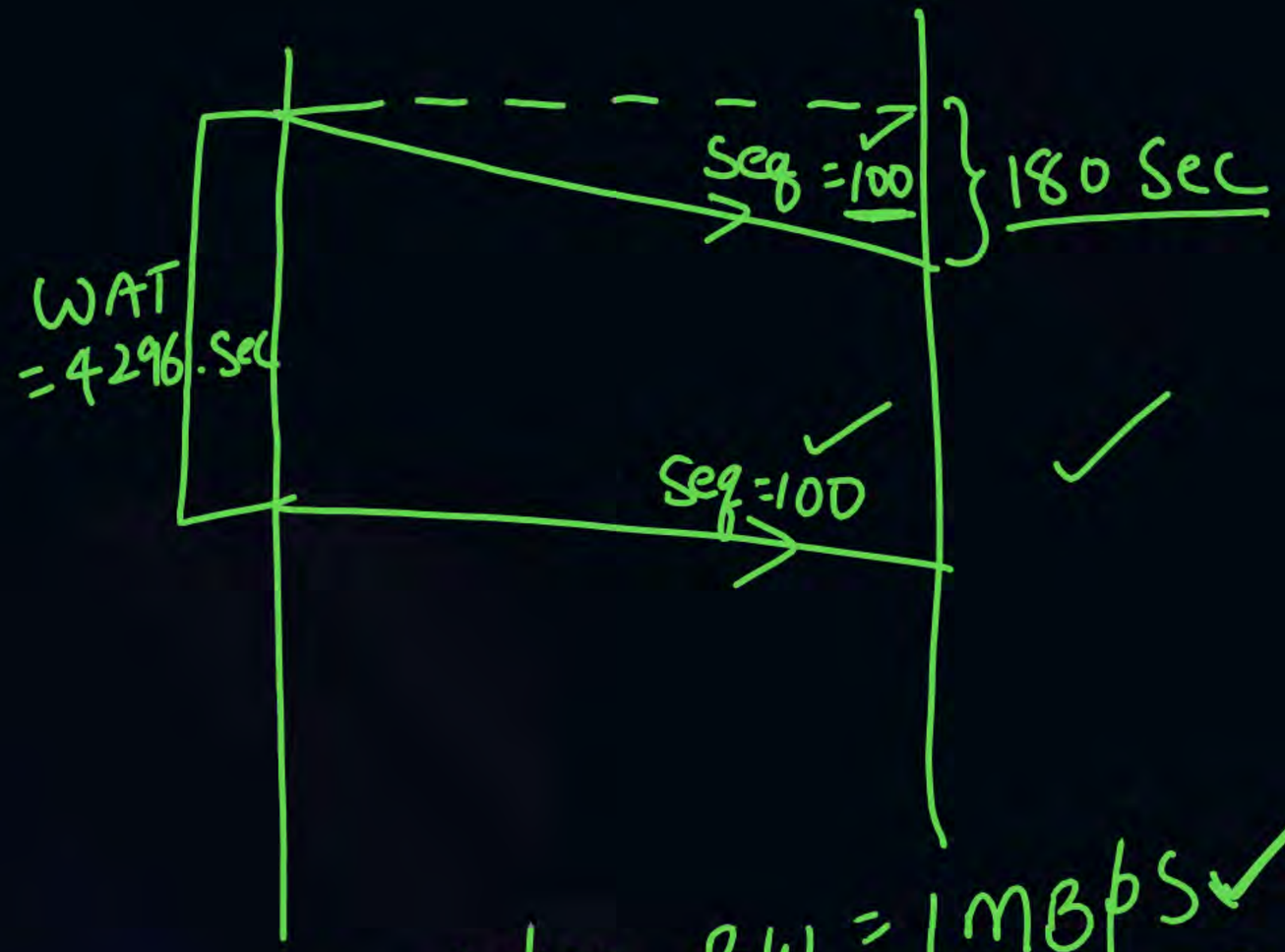
1 Seq num - $\frac{1}{10^6}$ sec

2^{32} Seq num — $\frac{2^{32}}{10^6}$ sec = $\frac{4294967296}{10^6}$ ✓

WAT = 4294.967296 sec

LT of a packet = 3 min = 180 sec

WAT > LT



when BW = 1 MBPS ✓

BW = 1 GBPS ✓

LT, WAT

THANK - YOU